Empirical Study of Memorization in Differentially Privately Fine-Tuned Large Language Models

Yuwei (Johnny) Meng

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Link to GitHub Repo

Abstract

1. Introduction

2. Background

Differential privacy (DP) is a mathematical framework that leverages probability theories to rigorously quantify privacy in algorithms (Dwork et al., 2006). Formally, let $\mathcal{X}$ denote the data space and $X = (x_1, \cdots, x_n) \in \mathcal{X}^n$ be a dataset. We can then define another dataset $X' = (x'_1, \cdots, x'_n) \in \mathcal{X}^n$ to be a *neighboring dataset* of X, denoted as $X \sim X'$, if there exists $i \in \{1, \cdots, n\}$ such that $x_i \neq x'_i$, and for all $j \neq i$, $x_j = x'_j$. With this setup, we say that a randomized algorithm $\mathcal{M}$ is (ε, δ)-*differentially private* if for all datasets $X \sim X'$, and all $S \subseteq \Omega$ where Ω is the output space of $\mathcal{M}$,

$$\mathbb{P}(\mathcal{M}(X) \in S) \leq e^{\varepsilon}\mathbb{P}(\mathcal{M}(X') \in S) + \delta.$$

In the context of training deep neural networks, we can adapt this definition and make gradient descent a differentially private algorithm so that the trained models do not memorize specific details of any training data. Defining $\mathcal{L}$ as the loss function, α as the learning rate, instead of updating the weights by $\alpha\nabla\mathcal{L}$ in every epoch, we first compute each per-sample gradient $\nabla\mathcal{L}_i$ and clip it to the norm bound C, then sum the clipped per-sample gradients and add random noise from $\mathcal{N}(0, \sigma^2 C^2 \boldsymbol{I})$ to the sum before updating. By choosing the parameters σ and C strategically, one can show that this algorithm, named *differentially private stochastic gradient descent* (DP-SGD), satisfies (ε, δ)-differential privacy (Abadi et al., 2016).

3. Related Work

4. Methodology

5. Results & Discussion

6. Conclusion

6.1. Limitations & Future Directions

Bibliography

[1] C. Dwork, F. McSherry, K. Nissim, and A. Smith, "Calibrating noise to sensitivity in private data analysis," in *Proceedings of the Third Conference on Theory of Cryptography*, in TCC'06. New York, NY: Springer-Verlag, 2006, pp. 265–284. doi: 10.1007/11681878_14.

[2] M. Abadi *et al.*, "Deep Learning with Differential Privacy," in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, in CCS '16. Vienna, Austria: Association for Computing Machinery, 2016, pp. 308–318. doi: 10.1145/2976749.2978318.

[3] A. Vaswani *et al.*, "Attention is All you Need," in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

[4] N. Carlini, C. Liu, Ú. Erlingsson, J. Kos, and D. Song, "The secret sharer: evaluating and testing unintended memorization in neural networks," in *Proceedings of the 28th USENIX Conference on Security Symposium*, in SEC'19. Santa Clara, CA, USA: USENIX Association, 2019, pp. 267–284.

[5] H. Du, S. Liu, and Y. Cao, "Can Differentially Private Fine-Tuning LLMs Protect Against Privacy Attacks?," in *Data and Applications Security and Privacy XXXIX*, S. Katsikas and B. Shafiq, Eds., Cham: Springer Nature Switzerland, 2025, pp. 311–329.